

9/15/26 Global/Local alignment

Suppose 2 sequences

Goal: Zinptseq

Output: Aligned seqs

GCG CATTGAGCGA

TGCGTTAGGGTACCA

need to find way to optimally align

indel

→ GCG CATTGAGCGA --

TGCG --TTAGGGTACCA

↑ mismatch match

Scoring Scheme

Unit Score	A	C	G	T	-
A	1	0	0	0	0
C	0	1	0	0	0
G	0	0	1	0	0
T	0	0	0	1	0
-	0	0	0	0	0

Alignment

A C G
1 1 1
A G G

$$Score = \sigma(A, A) + \sigma(C, G) + \sigma(G, G)$$

$$= 1 + 0 + 1 = 2$$

Gaps:

"-" is the gap symbol

$$\begin{array}{c} A A A G G G \\ G G G A A A \end{array} \rightarrow \begin{array}{c} - - - A A A G G G \\ G G G A A A - - - \end{array}$$

$$\text{Score} = 0 \qquad \text{Score} = 3$$

Dayhoff scores:

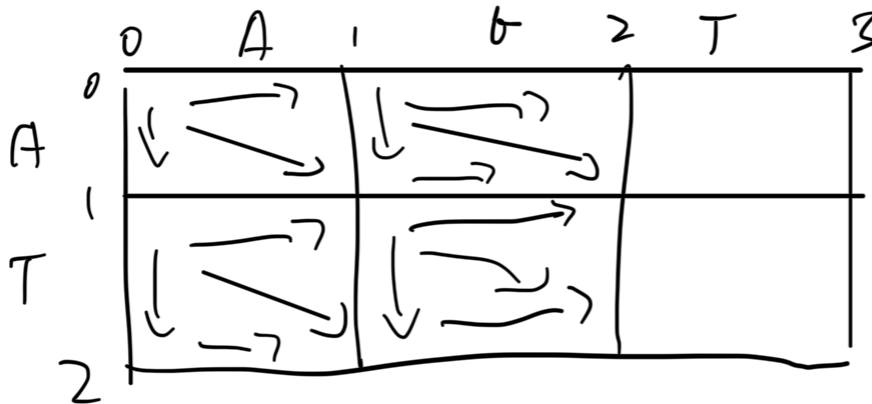
protein alignment sequences (a lot more sophisticated)

Scoring fxns

sum terms for pair of aligned residues
 - means: log likelihood of sequence existence

Edit Graph

alignment representation of all possible start and end nodes in the graph (alignment steps)



Edit graph has $(3+1) \cdot (2+1)$ nodes

Move to the right = $\rightarrow$
 move down = $\downarrow$

move diagonal $\hat{y}$

Every path from beginning to end is an alignment
Every alignment corresponds to path between beginning and end

The optimal answer to a problem is expressed in terms of optimal answers to its sub-problems

Given 2 seq X, Y and scoring scheme σ
Find an optimal alignment of X, Y

1. Compute 1st optimal alignment score
2. Construct optimal alignment

Optimal alignment is maximal score path in edit graph from begin vertex to end vertex

$S(i, j)$ = score of maximal cost path from begin vertex to end vertex

The optimal path to i, j must pass through $(i-1, j), (i, j-1), (i-1, j-1)$

This path val is $S(i-1, j) + \sigma(-, Y_j)$

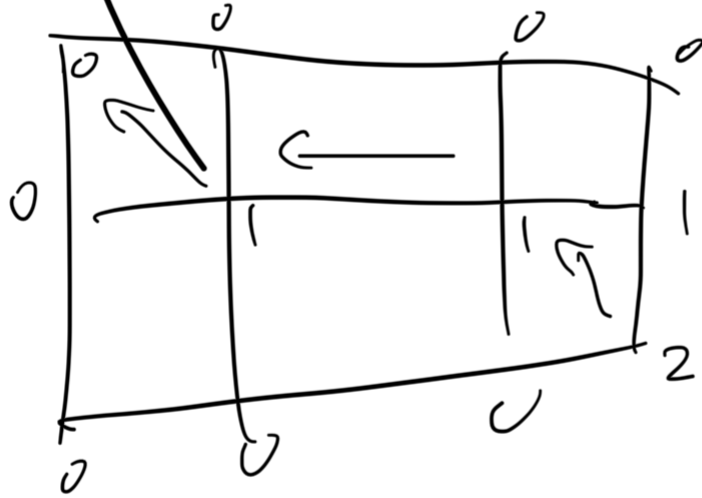
or $S(i-1, j-1) + \sigma(X_i, Y_j)$

or $S(i, j-1) + \sigma(X_i, -)$

Thus
$$S(i, j) = \max \begin{cases} S(i-1, j) + \sigma(-, Y_j) \\ S(i-1, j-1) + \sigma(X_i, Y_j) \\ S(i, j-1) + \sigma(X_i, -) \end{cases}$$

$$\{ s(i, j-1) + \sigma(x_i, -) \}$$

and unlabelled edges. Four DAG



Local Alignment

$$s(i, j) = \max \begin{cases} s(i-1, j) + \sigma(-, y_j) \\ s(i-1, j-1) + \sigma(x_i, y_j) \\ s(i, j-1) + \sigma(x_i, -) \\ 0 \end{cases}$$

General Scoring Schemes

Assume 1, independence of mutations

2. Gaps of any length are 1 mutation

Heuristic Interpretation of Alignment score

$X = A A C T A A$
 $Y = A C G G T G T$

Example

$$\begin{array}{ccccccc} A & A & C & G & - & T & (A) & (A) \\ A & - & C & G & G & T & (G) & (T) \end{array}$$

Let p be prob of match

let q be prob of sub/mismatch

let r be prob of single indel

$$P_{\text{alignment}} = p^4 q^2 r^2$$

Log likelihood

$s' = \log \text{prob of alignment}$

$$= 4 \log p + 2 \log q + 2 \log r$$

$$S = s' - 8 \log k = 4 \log p + 2 \log q + 2 \log r - 8 \log k$$

$$= 4 \cdot \log \frac{p}{k} + 2 \cdot \log \frac{q}{k} + 2 \cdot \log \frac{r}{k} \quad \alpha = \frac{k}{q}, \quad \gamma = \frac{k}{r}$$

choose k such that $\log\left(\frac{p}{k}\right) = 1$

$$\geq 4 - 2\alpha - 2$$